

Methods + Results Draft 1

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1 Methods

1.1 Ethics

Ethical approval was granted by the Ethics Review Board of the Faculty of Social and Behavioural Sciences of Utrecht University.

1.2 Data

Consistent with Kowall et al. (2025), data from waves 1-7 (excluding wave 3) of the Survey of Health, Ageing and Retirement in Europe (SHARE) were used. SHARE is a longitudinal panel study conducted across 28 European countries, collecting data on health, social, economic and environmental policies, since 2004.

All variables needed for the construction of the decision space (outcome, exposure, adjustment, exclusion) were selected from different datasets that were available per wave. Then, one cleaned data set was created per wave, where the selected variables were linked through an ID variable. Lastly, all wave data sets were combined based on the selected Data Type option in the multiverse analysis. After data cleaning, but before applying the exclusion criteria and missing data handling, the samples consisted of:

- cross-sectional all waves: 138,807 participants across 336,804 observations.
- cross-sectional wave 1: 30,416 participants across 30,416 observations.
- longitudinal no strict baseline: 87,029 participants across 285,026 observations.
- longitudinal wave 1: 23,301 participants across 91,935 observations.

1.3 Definition of decision space

To define the decision space, all options of six different decisions made by the teams in Kowall et al. (2025) were identified based on information provided in the original publication (see Table 1). When systematically combining all options across decisions, the decision space consisted of 129,024 universes. After removing invalid research paths, 82,944 valid universes remained. Invalid paths included combinations of cross-sectional data with models requiring longitudinal data structure (Cox proportional hazards (Cox PH), discrete-time mixed-effects), cross-sectional wave 1 data with models requiring repeated measures (mixed-effects logistic regression, generalised estimating equations (GEE)), wave 1 data with outcomes that were not available at baseline (since last interview, cause of death), and Cox models with outcomes lacking time-to-event variables.

1.4 Classifying sources of variation using the AB-framework

The selected decisions were classified using the AB-framework. Missing data and statistical model choice were classified as harmful sources of variation. Operationalisation of the outcome variable, selection of confounders, and exclusion criteria were classified as helpful. Data structure was also classified as helpful, as the framework treats variations over time as representing different target populations (see Table 1).

1.5 Converting OR and HR to RR

The statistical models yielded different effect size measures. Logistic regression, mixed-effects logistic regression, GEE, and discrete-time mixed-effects models produced odds ratios (OR). Cox PH models produced hazard ratios (HR). Log-binomial models produced risk ratios (RR) directly. To enable comparison across universes, all effect sizes were converted to RRs. Odds ratios and hazard ratios both approximate risk ratios when the outcome is rare, but can diverge substantially when the outcome is common (VanderWeele, 2020). Given that the overall prevalence in the unexposed group (p_0) was around 18% in all subsets of the data, conversions for both OR and HR were necessary.

Conversions to risk ratios often require knowing p_0 (Zhang & Yu, 1998). However, in a multiverse analysis, p_0 may vary across universes, depending on data type and exclusion decisions resulting in different data subsets. This could introduce arbitrary variation in the results, distorting the actual decision sensitivity.

Therefore, conversions proposed by VanderWeele (2020) were used, designed for scenarios when p_0 is unknown but exceeds 10%. For odds ratios (Huang et al., 2022a):

$$RR \approx \sqrt{OR} \tag{1}$$

Table 1: Analytical decisions and their options based on the original many-analyst study, that were used in the present multiverse analysis.

Decision	Option	Notes / Deviations from original
Harmful sources of variation		
Missing Data	Complete cases (16) “No information” (1)	For complete cases, listwise deletion was applied. One team replaced missing values in categorical variables by “no information” to prevent listwise deletion. In the original, some analysts used imputed data provided by SHARE, these were not used in the present multiverse analysis.
Statistical Model	Cox proportional hazards model (6) Logistic regression (5) Log-binomial model (2) Mixed-effects logistic regression (1) Discrete-time mixed-effects model (1) Generalised estimating equations (1)	The original study reported different effect sizes (OR/HR/RR). For comparability across universes, in this study, all effect sizes were converted to risk ratios (RR).
Helpful sources of variation		
Data Type	Cross-sectional wave 1 (5) Cross-sectional all waves (1) Longitudinal, wave 1 as only baseline (3) Longitudinal, no strict baseline (10)	The cleaned datasets for each wave were combined in different ways, depending on the selected option.
Exclusion Criteria	Subjects married before 18 Subjects from Israel Subjects under 55 at recruitment Subjects born after 1954 Change in exposure (widowed/divorced)	The analysts in the original study often used idiosyncratic exclusion criteria that were used in the present study. Because the original paper did not specify whether combinations of these exclusion criteria were used, a maximum of one exclusion criterion per universe was applied. *Note: one team removed all participants from Israel, as this is not an European country.
Outcome	Heart attack / stroke ever (16) Age at heart attack / stroke (9) Heart attack / stroke since last interview (7) Main cause of death: heart attack / stroke (5)	Seven combinations of the dependent variables were used by the analysts in the original study. Although other combinations are possible, the present study only included the seven operationalisations from the original study.
Adjustment Set	Age (12) Sex (12) Country (7) Smoking (6) Education (4) Physical activity (4) No confounders (2)	To ensure feasibility of the present multiverse analysis, only adjustment variables used by at least four analysts, except ‘no confounders’, were included to create the adjustment sets.

Note. Numbers in parentheses indicate how many analysts selected each option in the original study.

Note. For the exposure variable, all teams but two used “cohabiting married” (CM). Some analysts also included registered partnership, and “married living apart”. However, since the original research question only focused on CM, this was the only variable used as exposure in the present study.

For hazard ratios, where prevalence is estimated between 20% and 80% (Huang et al., 2022b; VanderWeele, 2020):

$$RR \approx \frac{1 - 0.5\sqrt{HR}}{1 - 0.5\sqrt{1/HR}} \quad (2)$$

Although the overall prevalence was slightly lower than 20%, a conversion not requiring a specific p_0 value was preferred to avoid introducing additional variation.

1.6 Round Robin sampling

Because the number of reasonable universes grows exponentially with the number of decisions and their options, the computational cost of multiverse analyses increases rapidly. To address this, Liu et al. (2023) proposed sampling universes rather than exploring the full decision space. They evaluated various algorithms and found that Round Robin sampling was one of the most effective methods for approximating multiverse results. Therefore, this method was applied in the present study. Round Robin sampling aims to represent each decision option at least once. The algorithm iteratively loops through all decisions and their options, sampling one universe at each stage without replacement, ensuring each sampled universe is included at most once. For a visual example of how the algorithm works, see Figure 1.

Although Liu et al. (2023) proposed sampling universes one by one and immediately processing the multiverse results, in this study universes were selected *a priori* before running the multiverse analysis. The sampler was implemented in two steps. First, decision options were mapped to corresponding row numbers in the full decision space. Then, row numbers were sampled according to the Round Robin algorithm and used to select universes from the original decision space. This approach was more computationally efficient than sampling directly from the decision space. A random seed was set to ensure reproducibility.

To ensure sufficient data were available for each option to quantify decision sensitivity, a minimum of 100 universes per option was sampled. As adjustment set had the most options (64), this resulted in 6,400 sampled universes in total.

Round Robin sampling algorithm

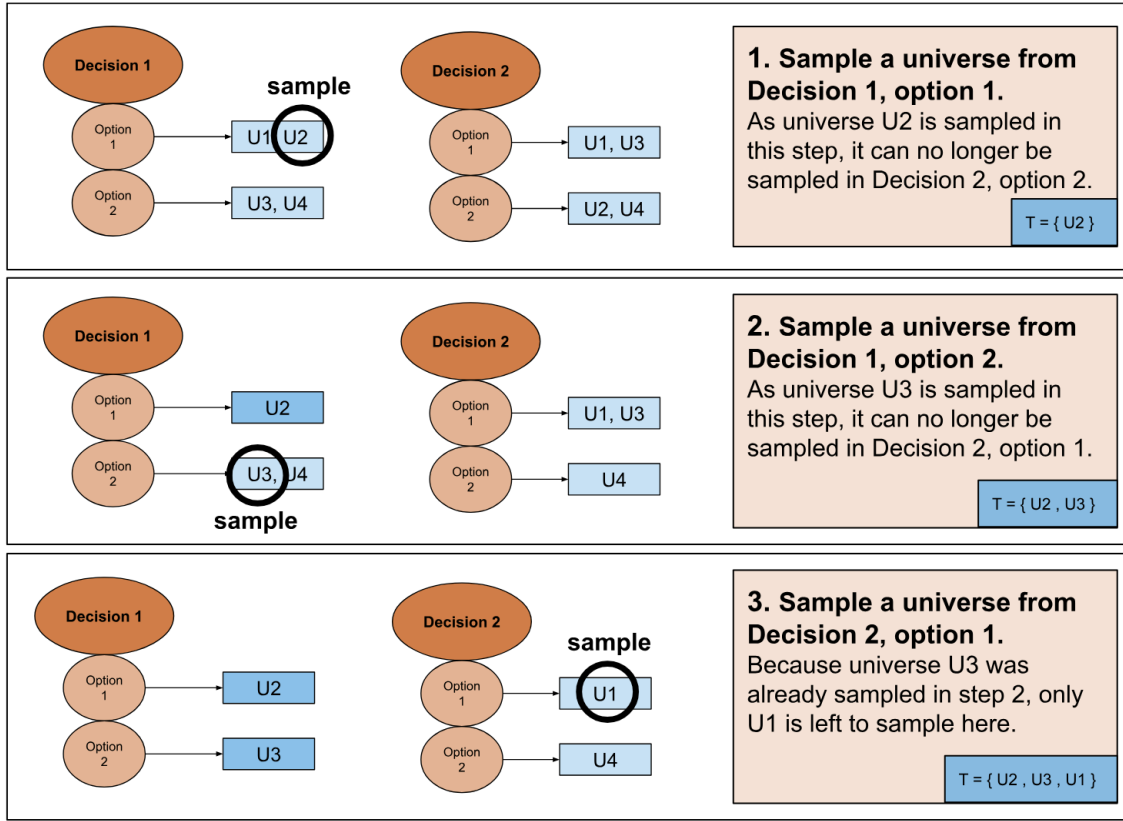


Figure 1: Schematic overview of Round Robin sampling algorithm. U: universe; T: the set of sampled universes. Figure was created by L. Ubbink.

1.7 Decision sensitivity

After converting all effect size estimates to risk ratios, density plots were used to visualise RR distributions across decision options. In addition, to visually assess the robustness of multiverse outcomes, each universe was plotted against its estimated effect size in a specification curve (Simonsohn et al., 2015). However, as both methods only provide descriptive information about the distribution of results rather than quantifying decision sensitivity, they were used as visual aids to inspect whether the outcome distribution was robust to different analytical specifications.

To quantify the decision sensitivity - the extent to which different options within each analytical decision lead to different outcomes - k-sample Anderson-Darling tests (AD tests) were used, as proposed by Liu et al. (2023). This test evaluates whether the RRs resulting from k options

within a decision stem from the same empirical cumulative distribution function (ECDF). If this null hypothesis was rejected, the decision was classified as sensitive. The AD statistic measures the weighted difference between the ECDFs of k samples. The standardised AD statistic (T.AD), which adjusts for number of groups and sample size, was used for between-decision comparisons. Higher T.AD values indicate more sensitive decisions.

Because there were 64 different adjustment sets, Kolmogorov-Smirnov tests (KS tests) were used for a more detailed evaluation of its decision sensitivity, comparing RR distributions for specifications including versus excluding each adjustment variable. The KS test statistic (D) represents the maximum absolute difference between the empirical cumulative distribution functions of the two samples, ranging from 0 (identical) to 1 (no overlap). Values closer to 0 indicate a higher likelihood that the two samples were drawn from the same distribution.

1.8 Supplementary Materials

All analyses were conducted in R (version 4.4.2, Team, 2024). All code scripts containing sampler and multiverse functions, and package versions documented in the lock file, can be found [here](#).

2 Results

2.1 Model convergence

Out of the 6,400 sampled universes, 1,487 failed to converge. Non-convergence mostly occurred in log-binomial models ($n = 1,102$), followed by generalised estimating equations (GEE; $n = 221$) and logistic regression ($n = 164$). The other three model types (Cox PH, mixed effects logistic, and discrete-time mixed effects) converged successfully in all cases. Across other analytical decisions, non-convergence was distributed approximately equally, suggesting these decisions did not substantially affect model convergence. In total, 4,913 universes yielded valid risk ratio estimates. The number of converged and non-converged universes per decision option can be found in Table 4.

2.2 Overall multiverse results

Across the 4,913 successful specifications, risk ratio estimates ranged from 0.71 to 1.68 (median = 0.95, mean = 0.94, SD = 0.1), with 61.1% of the universes yielding significant results ($p < .05$). Within this sample of the full multiverse, marriage thus appeared both as a protective factor ($RR < 1$) and a risk factor ($RR > 1$) for cardiovascular events, depending on the analytical specification. Figure 2 shows the full distribution of the effect estimates across universes, arranged from smallest to largest risk ratio, with significance indicated by colour.

Several patterns emerged in the specification curve (Figure 2). Universes including age as an adjustment variable were clustered around higher RRs (mostly ≥ 1), while universes without age were clustered around lower RRs (all < 1), suggesting a potential effect of age on the relationship between marriage and cardiovascular events. The highest RRs (> 1.25) only occurred in specifications including Cox PH models, although almost none of these estimates reached significance. Log-binomial models, in contrast, yielded only RRs below 1, almost all of which were significant. Furthermore, effect estimates seemed robust to data type (except cross-sectional W1), missing data handling, outcomes (except Age + Ever), and exclusion criteria, with no clear clustering patterns across these decisions.

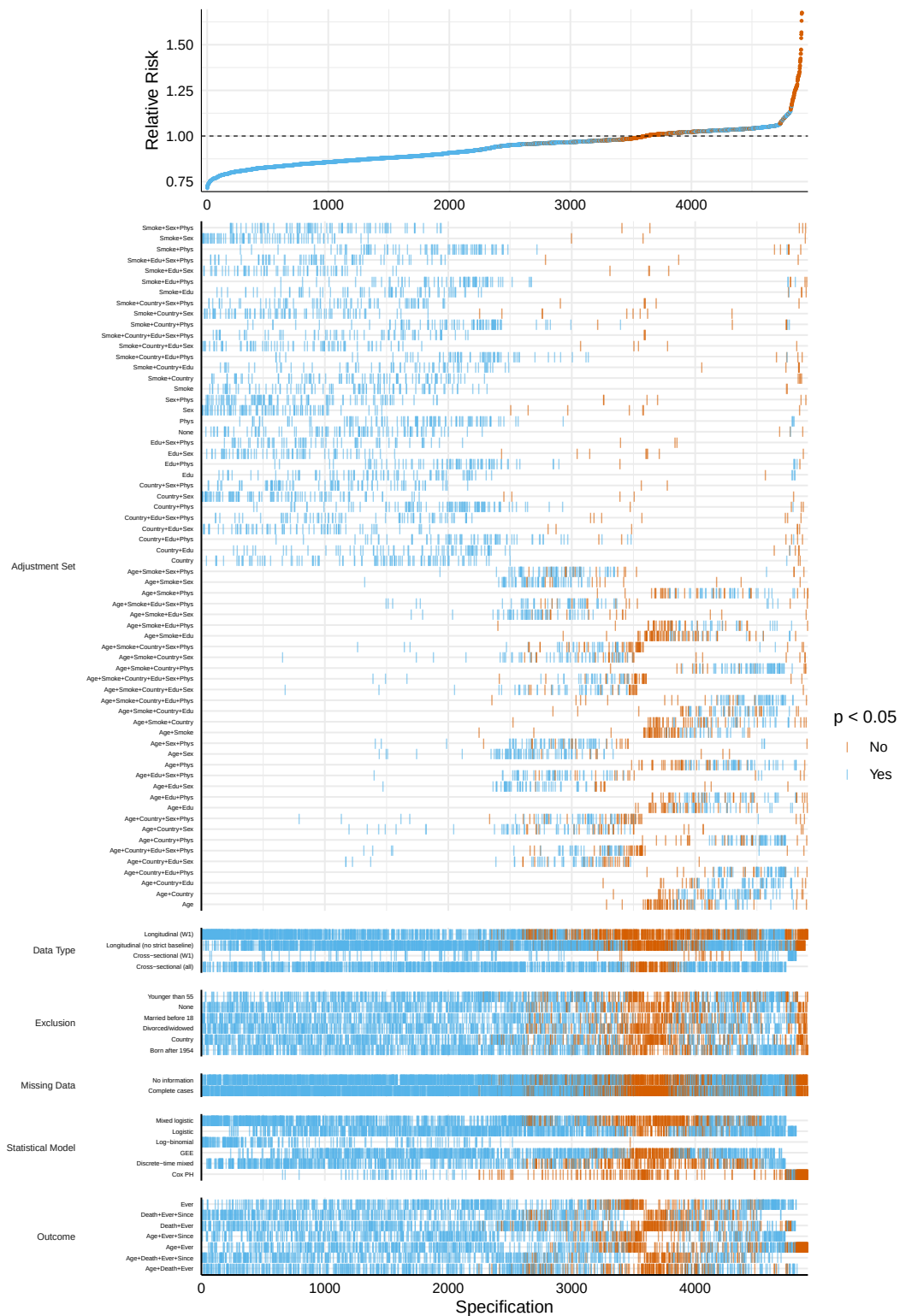


Figure 2: Specification curve multiverse results

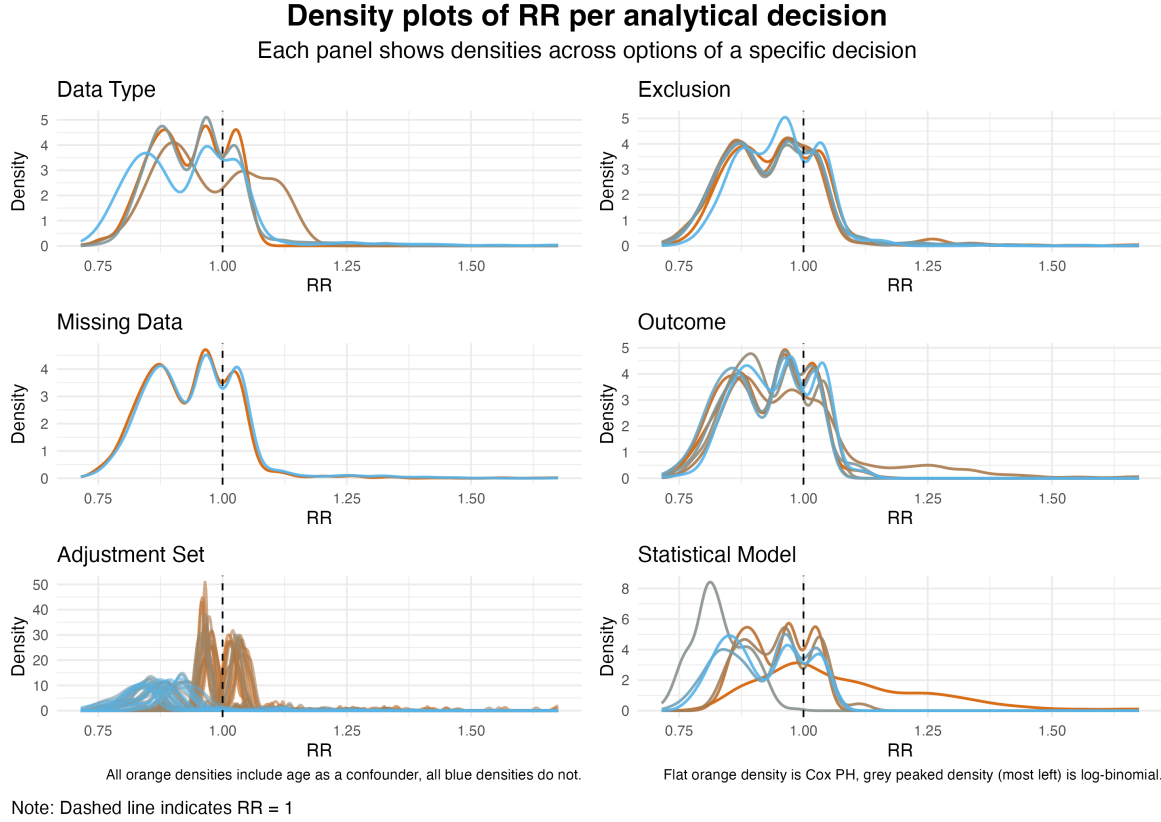


Figure 3: Densities of risk ratios (RR) across decisions and their options.

2.3 Decision sensitivity

To examine the effect size distributions across decision options in more detail, density plots were used (see Figure 3). Data type, missing data, exclusion criteria, and outcome showed largely overlapping RR distributions across their options. However, in case of statistical models and adjustment sets there were clear separations in RR distributions between options. Consistent with the pattern shown in Figure 2, log-binomial models were clustered below RR values of 1, while the density of Cox PH showed a long tail extending above 1. Similarly, specifications including age in the adjustment set showed a clear shift towards higher RRs compared to those without age as a confounder.

Mean RR ranges across options confirmed the patterns found in the specification curve and density plots (Table 4). Statistical model (0.83 - 1.08) and adjustment set (0.83 - 1.05) showed the largest variation in mean RR across options, while other decisions showed smaller ranges: exclusion criteria (0.94 - 0.94), missing data (0.93 - 0.95), data type (0.93 - 0.97), and outcome (0.92 - 0.98).

Table 2: K-sample Anderson-Darling tests for distributional differences across analytical decision options, with higher standardised AD values indicating more sensitive decisions.

Decision	k	T.AD	p-value
Missing Data	2	5.89	0.002
Exclusion Criteria	6	13.13	< .001
Data Type	4	50.88	< .001
Outcome	7	52.26	< .001
Statistical Model	6	307.10	< .001
Adjustment Set	64	394.79	< .001

Note. k = number of options; T.AD = standardised AD value.

A k-sample Anderson-Darling test (AD test) was used to quantify decision sensitivity (for the ECDF per decision see Figure 5). As shown in Table 2, AD tests for all decisions were highly significant (almost all $p < .001$), indicating distributional differences across options within each decision. Adjustment set and statistical model showed substantially higher T.AD values (307.1 and 394.79, respectively) than other decisions (missing data: 5.89; exclusion: 13.13; outcome: 50.88; data type: 52.26). This indicates that statistical model and adjustment set had the largest impact on the distribution of effect estimates. The number of options per decision did not seem to substantially affect the T.AD values.

Kolmogorov-Smirnov tests (KS test) were used to compare RR distributions for specifications including versus excluding each adjustment variable. For the ECDFs across adjustment variables, see Figure 6. As can be seen in Table 3, inclusion of all adjustment variables, except smoking, resulted in significantly different distributions than when these variables were excluded ($p < .001$). Age showed the largest effect (0.88), followed by not including any confounders (0.54) and sex (0.45). Physical activity, country, and education showed moderate, but significant, effects ($D_{activity} = 0.16$, $D_{country} = 0.1$, and $D_{education} = 0.08$). This confirms that the adjustment set sensitivity seems to be driven primarily by age, sex and the decision not to include confounders at all.

Table 3: Kolmogorov–Smirnov test for distributional differences across adjustment set options, when including or excluding specific adjustment variables. D: absolute max distance between the ECDFs of the two samples.

Adjustment variable	Included	Excluded	D	p-value
Smoking	2332	2581	0.03	0.341
Education	2214	2699	0.08	< .001
Country	2436	2477	0.10	< .001
Physical activity	2431	2482	0.16	< .001
Sex	2409	2504	0.45	< .001
None	100	4813	0.54	< .001
Age	2330	2583	0.88	< .001

Note. D = KS test statistic (range: 0 - 1), smaller numbers indicate smaller distributional differences.

The adjustment set sensitivity driven by age, sex and none is also visible in the RR density distributions (Figure 4). Including age as an adjustment variable shifted the RR distribution to the right (higher RRs), while including sex shifted it to the left (lower RRs). Specifications without any confounders also showed lower RRs. Other adjustment variables did not result in pronounced differences between RR distributions.

In the underlying data, married individuals were on average 4 years younger than unmarried individuals (65 versus 69 years old). Additionally, the distribution of sex differed between groups: 51% of married individuals were female, compared to 69% in the unmarried group.

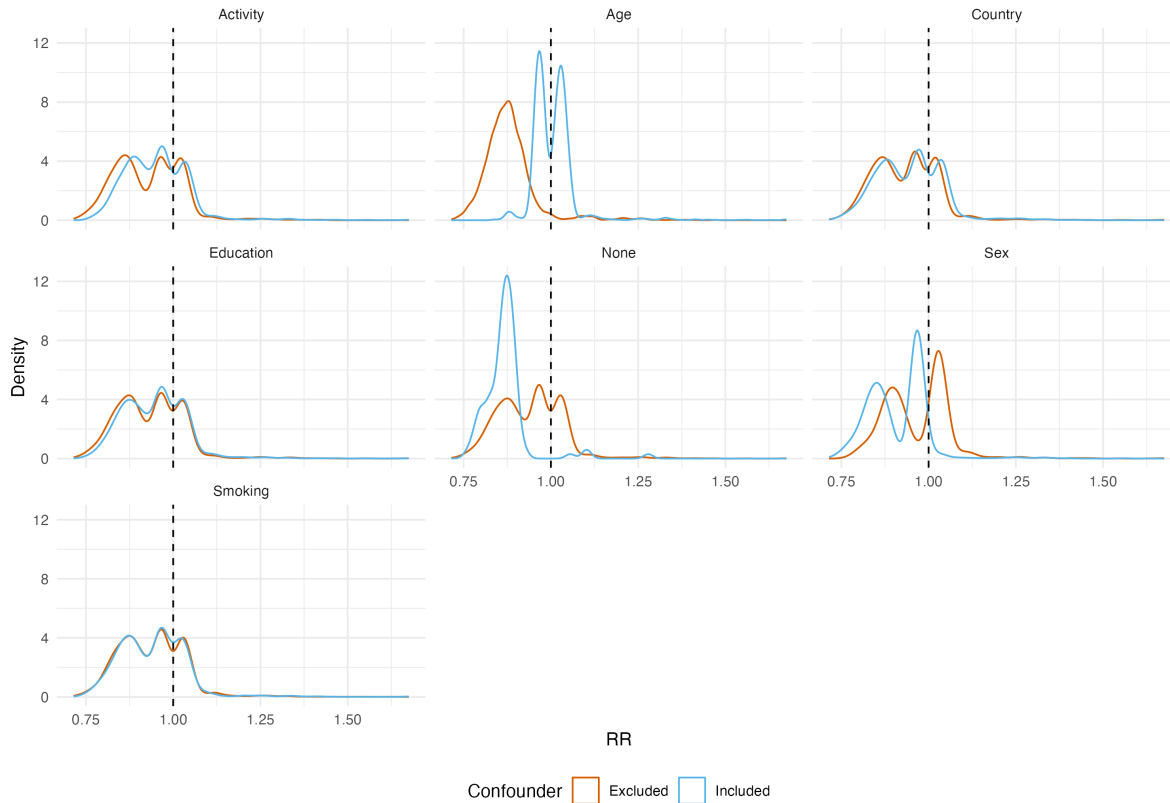


Figure 4: Densities of risk ratios (RR) across inclusion/exclusion of different adjustment variables.

3 References

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4 Appendix

Empirical cumulative distribution functions (ECDF) of RRs per analytical decision

Each panel shows ECDFs across decision options

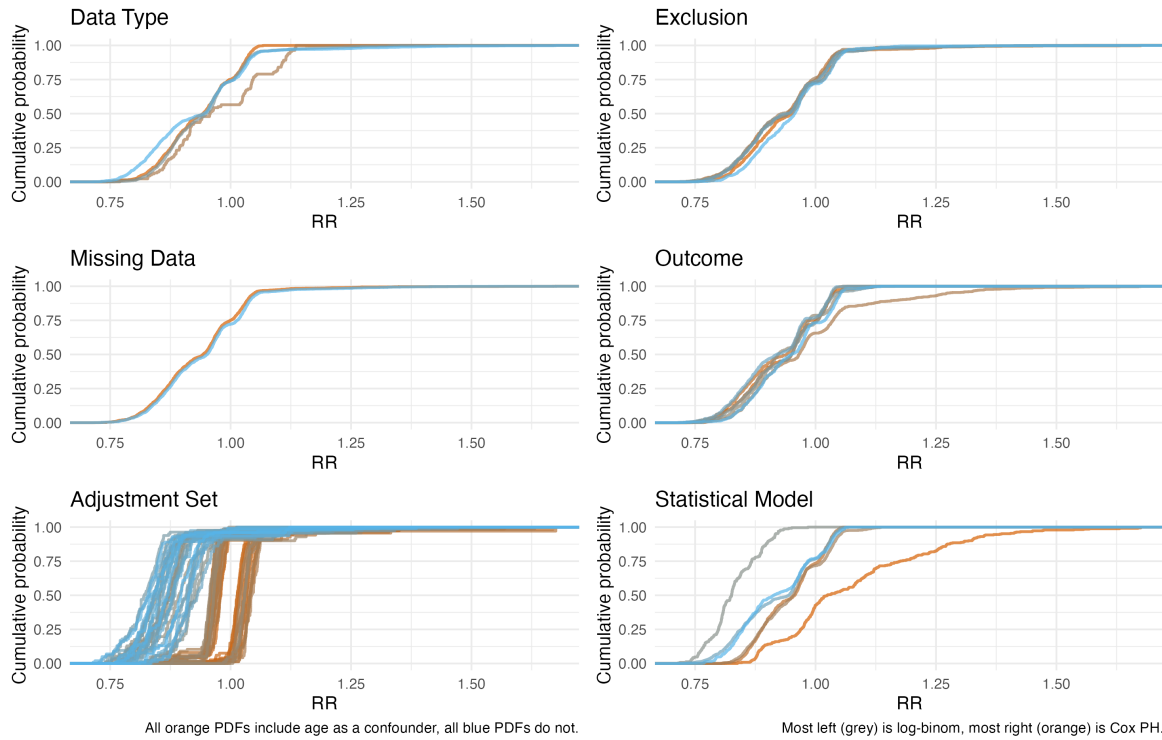


Figure 5: Empirical cumulative distribution functions of risk ratios (RR) across decisions and their options.

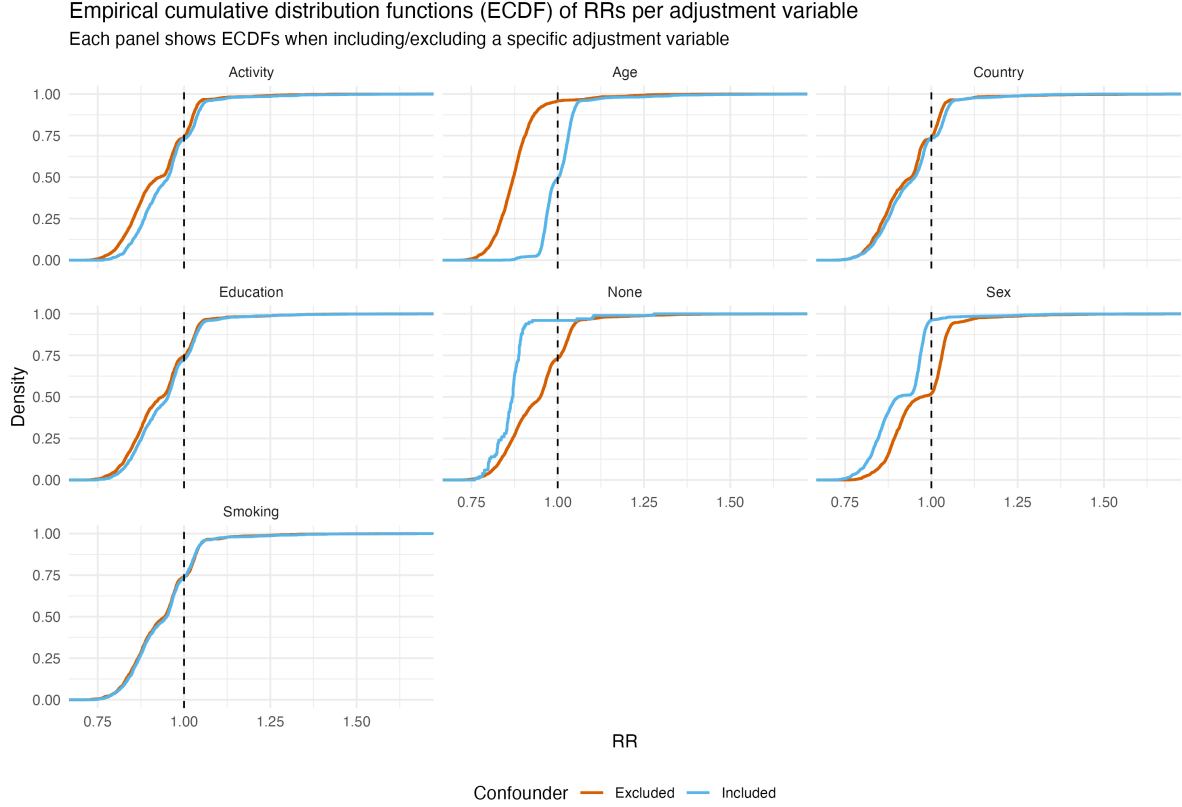


Figure 6: Empirical cumulative distribution functions of risk ratios (RR) across inclusion/exclusion of different adjustment variables.

Table 4: Number of universes including specific decision options, non-convergence, RRs, their range and percentage of significant RRs across decisions and their options.

Decision	Option	Successful Universes	Non- convergence	Mean RR	RR Range	% Sig
Statistical Model	Log-binomial	232	1102	0.83	[0.71, 0.98]	100%
Statistical Model	Mixed effects logistic regression	1278	0	0.92	[0.73, 1.07]	77%
Statistical Model	Discrete-time mixed effects	832	0	0.92	[0.76, 1.07]	74%
Statistical Model	GEE	1052	221	0.95	[0.8, 1.06]	85%
Statistical Model	Logistic regression	1188	164	0.95	[0.8, 1.14]	93%
Statistical Model	Cox PH	331	0	1.08	[0.84, 1.68]	24%
Data Type	Cross-sectional (all waves)	1237	403	0.93	[0.74, 1.07]	91%
Data Type	Longitudinal (wave 1)	1669	563	0.93	[0.71, 1.68]	69%
Data Type	Longitudinal (no strict baseline)	1869	411	0.95	[0.75, 1.35]	82%
Data Type	Cross-sectional (wave 1)	138	110	0.97	[0.77, 1.14]	86%
Missing Data	Complete cases	2584	637	0.94	[0.72, 1.68]	80%
Missing Data	No information	2329	850	0.94	[0.71, 1.68]	80%
Exclusion Criteria	Country	833	261	0.93	[0.73, 1.54]	76%
Exclusion Criteria	Divorced/widowed	819	277	0.93	[0.73, 1.47]	79%
Exclusion Criteria	Married before 18	778	248	0.94	[0.71, 1.45]	82%
Exclusion Criteria	None	804	263	0.94	[0.73, 1.56]	80%
Exclusion Criteria	Born after 1954	886	209	0.95	[0.74, 1.68]	83%

(continued)

Decision	Option	Successful Universes	Non- convergence	Mean RR	RR Range	% Sig
Exclusion Criteria	Younger than 55	793	229	0.95	[0.76, 1.67]	79%
Outcome	Age event + Death + Event ever + Since	644	193	0.92	[0.73, 1.06]	84%
Outcome	Death + Event ever + Since	656	182	0.92	[0.71, 1.06]	84%
Outcome	Age event + Death + Event ever	707	177	0.93	[0.75, 1.16]	76%
Outcome	Age event + Event ever + Since	617	216	0.93	[0.74, 1.07]	86%
Outcome	Death + Event ever	761	187	0.93	[0.72, 1.13]	76%
Outcome	Event ever	695	266	0.95	[0.73, 1.14]	83%
Outcome	Age event + Event ever	833	266	0.98	[0.77, 1.68]	72%
Adjustment Set	Smoke + Sex	81	19	0.83	[0.74, 1.35]	97%
Adjustment Set	Country + Sex	100	0	0.83	[0.73, 1.12]	95%
Adjustment Set	Sex	100	0	0.83	[0.71, 1.22]	93%
Adjustment Set	Smoke + Edu + Sex	65	35	0.84	[0.75, 1.01]	93%
Adjustment Set	Smoke + Country + Edu + Sex	74	26	0.85	[0.74, 1.16]	94%
Adjustment Set	Smoke + Country + Sex	85	15	0.85	[0.75, 1.03]	90%
Adjustment Set	Edu + Sex	75	25	0.85	[0.76, 1.31]	90%
Adjustment Set	Sex + Phys	100	0	0.85	[0.76, 1.39]	94%
Adjustment Set	Smoke	86	14	0.86	[0.77, 1.08]	98%
Adjustment Set	Smoke + Edu + Sex + Phys	66	34	0.86	[0.77, 1.2]	96%
Adjustment Set	Country + Edu + Sex	73	27	0.86	[0.75, 1.22]	91%
Adjustment Set	Country + Sex + Phys	79	21	0.86	[0.77, 1.01]	97%
Adjustment Set	Edu + Sex + Phys	68	32	0.86	[0.8, 1.02]	96%
Adjustment Set	Smoke + Country + Edu + Sex + Phys	63	37	0.87	[0.78, 1]	93%
Adjustment Set	Smoke + Country + Sex + Phys	72	28	0.87	[0.77, 1.21]	92%
Adjustment Set	Smoke + Sex + Phys	84	16	0.87	[0.8, 1.38]	94%
Adjustment Set	None	100	0	0.87	[0.76, 1.28]	97%
Adjustment Set	Smoke + Country	87	13	0.88	[0.78, 1.26]	96%
Adjustment Set	Country	100	0	0.88	[0.76, 1.24]	98%
Adjustment Set	Country + Edu + Sex + Phys	73	27	0.89	[0.79, 1.25]	91%
Adjustment Set	Edu	78	22	0.89	[0.79, 1.13]	99%
Adjustment Set	Smoke + Country + Edu	66	34	0.90	[0.79, 1.21]	93%
Adjustment Set	Smoke + Edu	67	33	0.90	[0.79, 1.41]	95%
Adjustment Set	Phys	100	0	0.90	[0.81, 1.12]	100%
Adjustment Set	Country + Edu	75	25	0.91	[0.81, 1.21]	93%
Adjustment Set	Smoke + Country + Phys	84	16	0.92	[0.82, 1.1]	94%
Adjustment Set	Smoke + Phys	87	13	0.92	[0.81, 1.25]	93%
Adjustment Set	Country + Phys	100	0	0.92	[0.82, 1.31]	96%
Adjustment Set	Edu + Phys	79	21	0.93	[0.83, 1.24]	98%
Adjustment Set	Smoke + Country + Edu + Phys	69	31	0.94	[0.83, 1.27]	95%
Adjustment Set	Smoke + Edu + Phys	75	25	0.94	[0.84, 1.47]	95%
Adjustment Set	Country + Edu + Phys	72	28	0.94	[0.83, 1.27]	92%
Adjustment Set	Age + Smoke + Sex	74	26	0.96	[0.87, 1.24]	82%
Adjustment Set	Age + Country + Edu + Sex	69	31	0.96	[0.87, 1.02]	63%
Adjustment Set	Age + Edu + Sex	60	40	0.96	[0.88, 1.09]	80%
Adjustment Set	Age + Sex	82	18	0.96	[0.88, 1.03]	88%
Adjustment Set	Age + Smoke + Country + Edu + Sex	62	38	0.97	[0.84, 1.3]	69%
Adjustment Set	Age + Smoke + Edu + Sex	68	32	0.97	[0.89, 1.41]	79%
Adjustment Set	Age + Smoke + Sex + Phys	79	21	0.97	[0.9, 1.33]	77%
Adjustment Set	Age + Country + Sex	77	23	0.97	[0.87, 1.28]	68%
Adjustment Set	Age + Sex + Phys	79	21	0.97	[0.88, 1.56]	78%
Adjustment Set	Age + Smoke + Country + Edu + Sex + Phys	61	39	0.98	[0.88, 1.26]	54%
Adjustment Set	Age + Smoke + Country + Sex	75	25	0.98	[0.84, 1.35]	65%
Adjustment Set	Age + Smoke + Edu + Sex + Phys	62	38	0.98	[0.88, 1.57]	80%
Adjustment Set	Age + Country + Edu + Sex + Phys	80	20	0.98	[0.87, 1.33]	57%
Adjustment Set	Age + Country + Sex + Phys	86	14	0.98	[0.85, 1.33]	57%
Adjustment Set	Age + Edu + Sex + Phys	65	35	0.98	[0.88, 1.33]	76%
Adjustment Set	Age + Smoke + Country + Sex + Phys	72	28	0.99	[0.87, 1.32]	43%
Adjustment Set	Age + Smoke	75	25	1.01	[0.92, 1.04]	37%
Adjustment Set	Age	87	13	1.02	[0.98, 1.42]	33%
Adjustment Set	Age + Smoke + Edu	80	20	1.02	[0.96, 1.13]	27%
Adjustment Set	Age + Smoke + Edu + Phys	61	39	1.03	[0.94, 1.67]	51%
Adjustment Set	Age + Smoke + Country	73	27	1.04	[0.95, 1.42]	44%
Adjustment Set	Age + Smoke + Country + Edu	67	33	1.04	[0.96, 1.2]	60%
Adjustment Set	Age + Smoke + Country + Edu + Phys	66	34	1.04	[0.94, 1.45]	77%

(continued)

Decision	Option	Successful Universes	Non- convergence	Mean RR	RR Range	% Sig
Adjustment Set	Age + Country	87	13	1.04	[0.98, 1.28]	74%
Adjustment Set	Age + Edu	71	29	1.04	[0.99, 1.63]	48%
Adjustment Set	Age + Phys	85	15	1.04	[0.98, 1.34]	60%
Adjustment Set	Age + Smoke + Country + Phys	78	22	1.05	[0.98, 1.4]	82%
Adjustment Set	Age + Smoke + Phys	68	32	1.05	[0.97, 1.68]	58%
Adjustment Set	Age + Country + Edu	63	37	1.05	[0.98, 1.37]	75%
Adjustment Set	Age + Country + Edu + Phys	71	29	1.05	[1, 1.47]	78%
Adjustment Set	Age + Country + Phys	77	23	1.05	[0.99, 1.45]	70%
Adjustment Set	Age + Edu + Phys	70	30	1.05	[1, 1.54]	59%